Model Sample Paper I

Computer Applications

Class IX

Time: 2 hrs. Marks: 100

Answer all questions from 'Section A' and any four from 'Section B'.

SECTION A (40 Marks)

Q.1. (a) Name two jump statements used in Java programming.

(b) Name the type of error in each case:

(i) Division by a variable that contains a value of zero.

(ii) Multiplication operator used when the operation should be division.

(c) Explain Data Abstraction with an example.

(d) What is a Package? Give an example. [2+2+2+2]

Q.2. (a) Explain if-else-if construct.

(b) What is a compound statement? Give an example.

(c) Define:

(i) Ternary Operator

(ii) Implicit Conversion [2+2+4]

Q.3. Give two differences between:

(a) Compiler and Interpreter

(b) Syntax Error and Logical Error [8]

Q.4. Write down the following expressions in Java:

(a) $\frac{\sqrt{3}}{4}a^2$ (b) $a^2 + b^3 + c^4$

(c) $\frac{(a+b)^2}{ab}$ (d) $\sqrt[3]{mn} + mn^3$ [8]

Q.5. (a) Rewrite the snippet, using Ternary operators:

```
if(a<b)
 {
   c=(a+b);
 {
else
 {
   c=(a-b);
 }
```

(b) Predict the output:

```
int a=6,b=5;
a+ = a++ % b++ *a + b++* --b;
```

(c) Give the output of the snippet:

```
int a=10,b=12;
if(a==10&&b<=12)
  a--;
else
  ++b;
System.out.println(a + "and" +b);
```

(d) Give the output of the snippet:

```
int p = 9;
while (a<=15)
{
p++;
if(p== 10)
   continue;
System.out.println(p);
}
```

[2+2+2+2]

SECTION B (60 Marks)

(15 × 4 = 60)

Answer any four:

Q.6. An Electricity Board charges for electricity per month from their consumers according to the units consumed. The tariff is given below:

Units Consumed	Charges
Up to 200 units	₹ 3.80/unit
More than 200 units and up to 300 units	₹ 4.40/unit
More than 300 units and up to 400 units	₹ 5.10/unit
More than 400 units	₹ 5.80/unit

Write a program to calculate the electricity bill taking consumer's name and units consumed as inputs. Display the output.

Q.7. The equivalent resistance of a series and a parallel connection of two resistances are given by the formula:

$R_1 = r_1 + r_2$

$R_2 = \frac{r_1 . r_2}{r_1 + r_2}$

Write a program to input the value of r_1 and r_2. Calculate and display the output of the equivalent resistance as per the user's choice.

Q.8. (a) Write a program to display the first 10 terms of the series:
5,10,17, ...

(b) Write a program to display the given pattern:

```
1 1 1 1 1
3 3 3 3
5 5 5
7 7
9
```

Q.9. (a) Write a program to find and display the sum of first 20 terms of the series:

$S = \frac{2}{3} + \frac{4}{5} + \frac{8}{7} + \frac{16}{9} + \ldots\ldots\ldots\ldots$

(b) Write a program to display the given pattern:

```
3
5  6
8  9  10
12 13 14 15
```

Q.10. Write a program to input a number. Find the sum of its digits and the number of digits. Display the output.

Sample Input: 7359
Sum of digits = 24
Number of digits = 4

Practice Oriented Competency Focused Questions

Computer Applications

Maximum Marks: 50

Class: IX

Time allowed: One hour (inclusive of reading time)
All questions are compulsory.

SECTION A (20 Marks)

I. Choose the correct answer: [1 × 15]

1. The ……………… is called an instance of a class.
   (a) Object (b) Attributes (c) State (d) None of these
2. The ……………… allows a class to use the properties and methods of another class.
   (a) Inheritance (b) Polymorphism
   (c) Encapsulation (d) None of these
3. Which is not an Object Oriented Programming language?
   (a) Python (b) Eiffel (c) Small talk (d) COBOL
4. Which software company produced Java?
   (a) Apple (b) Sun Microsoft (c) Samsung (d) Microsoft
5. Which keyword includes a package in a Java program?
   (a) New (b) public (c) import (d) class
5. The number of bytes occupied by char data type is ………..bytes.
   (a) 4 (b) 8 (c) 2 (d) 6
6. The memory capacity (storage size) of short and float data types are ……………. bytes.
   (a) 1 and 2 (b) 2 and 3 (c) 2 and 4 (d) 2 and 6
7. The ……………… are the words which have special meaning in Java programming.
   (a) keywords (b) identifiers (c) methods (d) package
8. Arrange the operators given below in order of higher precedence to lower Precedence:
   (i) && (ii) % (iii) >= (iv) ++
   (a) ++, &&, >=, % (b) ++, >=, %, &&
   (c) ++, %, >=, && (d) &&, %, >=, ++
9. Which of the following are valid comments?
   (a) /* comment */ (b) /* comment//
   (c) // comment (d) *// comment */
10. Which of the following keywords is used to create an object of a class?
    (a) new (b) public (c) class (d) None of these

11. Name the type of error in the statement given below:
int p =99/0;
(a) Syntax error (b) Runtime error (c) Logical error (d) None of these

12. If the value of basic=1500, what will be the value of tax after the following statement is executed?
tax = basic>1200? 200 :100;
(a) 100 (b) 200 (c) 300 (d) 400

13. Predict the output of Math.sqrt(Math.max(9,16)).
(a) 3.0 (b) 4.0 (c) 5.0 (d) 2.0

14. What is the final value stored in variable p?
double a = – 9.45;
double p = Math.abs(Math.floor(a));
(a) 9.0 (b) 8.0 (c) 10.0 (d) 9.5

15. What will be the final value of the expression, if x=2, y=1 and z=3?
p = x+ --y + z++ + y
(a) 5 (b) 6 (c) 4 (d) 7

II. Choose the odd one out from the following: [5]

(a) (i) Encapsulation (ii) Data Abstraction
(iii) Compiler (iv) Polymorphism
(b) (i) > (ii) ==
(iii) && (iv) <
(c) (i) case (ii) break
(iii) default (iv) System.exit(0)
(d) (i) int (ii) double
(iii) char (iv) String
(e) (i) + (ii) %
(iii) / (iv) ||

Section: B (30 marks)

III. Assertion and Reasoning: [2x4]

While answering these questions, you must be aware that the assertion is the fact and reason is the description of the assertion. Based on the assertion and its reason, you need to pick an appropriate option from a number of options.

1. Assertion (A): Java language is case sensitive.
Reason (R): The case sensitive means, it differentiates between the uppercase and lowercase letters by the language.
Based on the above assertion and reasoning, pick an appropriate statement from the options given below:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true and R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.
(e) Both A and R are false.

2. Assertion (A): Keywords are the reserved words which are preserved by Java language.
Reason (R): The keywords do not carry any special meaning for the language translator. So, they can be used as variable names.

Based on the above assertion and reasoning, pick an appropriate statement from the options given below:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true and R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.
(e) Both A and R are false.

3. Assertion (A): In Java, the operators (/) and (%) are known as division operators.

Reason (R): The operator (/) provides the result after dividing one operand by the other. Whereas, the (%) operator results in the remainder after dividing one operand by the other.

Based on the above assertion and reasoning, pick an appropriate statement from the options given below:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true and R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.
(e) Both A and R are false.

4. Assertion (A): Comments are non-executable statements that enable the users to understand the program logic.

Reason (R): They are basically remark statements. The comments are used to explain the code and to make it more informative for the users.

Based on the above assertion and reasoning, pick an appropriate statement from the options given below:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true and R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.
(e) Both A and R are false.

III. Give the output of following snippets: [2x5]

1.
```
if (a > b)
    System.out.println(a+b);
    System.out.println (a*b);
when a = 4 and b = 6
```

2.
```
int b=3,b,r;
float a=15.15,c=0;
if (p==1)
{
   r=(int)a/b;
   System.out.println(r);
}
else
{
   c=a/b;
   System.out.println(c);
}
```

3. int i;

```
for( i=5 ; i>=1 ;i--)
{
  if(i%2 ==1)
    continue;
  System.out.print( i+ " ");
}
```

4. When k='B'

```
switch (k)
{
  case 'a':
    System.out.println("Computer");
  case 'b':
    System.out.println("Science");
    break;
  case 'c':
    System.out.println("English");
  default:
    System.out.println("Activities");
}
```

5. int n=1000;

```
while (n>10)
{
  n=n/10;
}
System.out.println(n);
```

IV. Application based questions: (4x3)

1. The two decision control statements in Java are *if* and *switch*. The decision control statements are used to check for condition and execute the statements based on the condition. Further, an *if* statement within another *if* statement is termed as *nested if* statement. The switch is called as multiple branching statement. In Java programming, we also use the repetitive execution of a set of statements, termed as looping. The two types of loops are entry controlled and exit controlled loops. The *while* and *for* are termed as entry-controlled loops. A *for* loop is used when the number of iterations is known. The *while* loop is used when the number of iterations is not known and the set of statements are executed as long as the condition is true.

   Based on the above discussion, answer the following questions:

   (*a*) Which of the following refers to as the decision control statements in Java?
   - (*i*) if and switch
   - (*ii*) for and while
   - (*iii*) if and for
   - (*iv*) None of these

   (*b*) When an *if* statement is used within another *if* statement, it is termed as:
   - (*i*) Nested loop
   - (*ii*) Nested while
   - (*iii*) Nested if
   - (*iv*) Nested for

(c) What is the appropriate name given for the repetitive execution of a set of statements?

(i) Looping (ii) Decision control

(iii) Assignment (iv) Execution control

(d) Which of the following is/are known as an entry-controlled loop?

(i) for loop (ii) while loop

(iii) do-while loop (iv) All of these

2. In Java, there are various functions used for mathematical operations. The application of these functions in a program enhances the capability of processing and also reduces the complexity of the programming logic. Name the function for each of the statements given below:

(a) This function is used to return an absolute value (i.e., only magnitude of the number).

(b) This function returns the higher integer between the two consecutive integers, in which the given argument falls.

(c) It will generate a double type number between 0 and 1.

(d) This function is used to find the square root of a positive number.

3. As a homework assignment, the teacher has given a code to generate a pattern as shown along right side. There are some places in the code marked with ?1?, ?2?, ?3? and ?4? to be filled with data type statement/expression so that the code should execute properly.

The code is as shown below:

Homework assignment

```
# To create a pattern
..?1?.. i, j;
for (i=5; i>=1; ..?2?..)
{
    for (..?3?..; j>=1; j--)
    {
        System.out.print("*");
    }
    ............?4?...............
}
```

```
*****
****
***
**
*
```

Now, answer the following questions:

(a) What will be the data type in place of ?1?

(b) What will be the statement/expression in place of ?2?

(c) What will be the statement/expression in place of ?3?

(d) What will be the statement/expression in place of ?4?